

Examiner reports

Q1.

Surprisingly, only just over half of students got part (a) correct. The most frequent errors seen were numerical slips, writing -4 when the question asked for magnitude and picking the wrong section of the graph. Most seemed to know they had to find a gradient.

Just under half of students scored full marks on part (b). Of the students who knew they had to compare areas above and below the time axis, several used correct but elaborate methods often setting up equations involving time. The most successful students simply counted squares often showing any working on the diagram. Common errors included working with gradients instead of areas or counting incorrect sections of the graph as positive areas when they should have been negative.

Q8.

There were many good attempts at this question. In part (a)(i), many candidates were able to obtain the correct answer although some found the time and then the acceleration.

There were only a few cases of candidates confusing u and v in their equations. In part (a)(ii), there were again many good responses, but in some cases inappropriate information was used, for example using a distance of 75m or a final speed of 10m s^{-1} . In part (a)(iii) there were again many correct answers, but quite a few candidates did not find the magnitude of the force and gave negative answers. This was also the case in part (b), where candidates also sometimes added the 200 to their previous answer instead of subtracting it from it.

Q9.

There were some very varied responses to this question. Part (a) was often done well and the candidates who attempted to find the position vector generally gained the marks available. However, quite a few gave the velocity instead of the position vector. Those who had a velocity vector in part (a) then generally used this in part (b) and scored no marks. Some who had a correct position vector in part (a), used the wrong component to try to answer part (b).

Part (c) was altogether more challenging and many candidates worked with position vectors rather than velocity vectors. This clearly was the wrong approach, but some expended a lot of effort solving the quadratic equations that they produced. Some candidates came up with the correct velocity, but without showing how they had obtained their answer. It is important that working is justified. Sometimes a trial and improvement method led candidates to the correct answer.

Q10.

There were many correct answers to part (a) and the candidates were clearly helped by the presence of the printed answer. In most cases the arguments given were sound, but in others this was not the case. For example some candidates simply wrote statements such as $22.4 \sin \theta - 19.6$ and never included an equals sign or a zero. This was regarded as an important omission.

There were also lots of good answers to part (b), but some basic errors, such as the omission of a negative sign with g or the use of $\sin(0.875)$. In part (c) there were also some good answers. Some candidates did not realise that the time of flight could be found by doubling the 2 seconds that was given, and did quite a lot of work finding the time of flight. In some cases there were errors in the calculation of the distance. For example an

incorrect application of $s = \frac{1}{2} (u + v) t$ was seen with statements such as $s = \frac{1}{2} (10.8 + 0) \times 4$.

The approaches to part (d) were many and varied. There were quite a number of good responses, but often candidates did not really get very far with this part of the question. The most common error was to see candidates use a distance for example $19.6 - 5 = 14.6$, with the initial speed of $22.4 \sin \theta$.

In the final part, there was a tendency for the candidates to find the resultant of two velocity components, as they did not realise that for the minimum speed the vertical component would be zero. In some case the correct final answer was obtained from expressions such as $\sqrt{10.8^2 + 0^2}$.

Q11.

This question was also answered well by virtually all candidates, who showed that they knew the techniques involved in answering it. In part (c) the majority of students integrated to find the position vector but a common error was to forget the $+ c$ term and thus to ignore the initial position vector. Some found the $+ c$ incorrectly assuming that c was zero without completing the necessary calculation.

Q12.

This question was also done very well by the candidates. The main issues were in parts (a) and (b). In part (a), a number of candidates made errors when calculating the area under the graph. Often they would have one small error, which meant that their total was incorrect. In these cases, the candidates' distances were followed through into part (b). Both parts (b) and (c) were done well with very few candidates losing marks. In part (d), Newton's second law was applied in all but a few cases, but the issue was with the units of the mass. Some candidates converted to kg incorrectly, while others simply found the force as if the mass were 200 kg.

Q13.

Parts (a) and (b) were generally done well, with the candidates gaining many marks. A few candidates could not state the direction in part (b)(ii) and gave answers such as north-east or south-west. Part (c) was found to be more demanding. Candidates who did not identify the $\mathbf{i}$ and $\mathbf{j}$ components of the velocity made very little progress. Some made some initial good progress, but made arithmetic or algebraic errors in their solutions. Some forgot to square the speed and tried to work from equations such as $(4 - 0.4t)^2 + (0.5 + 0.2t)^2 = 5$. Some candidates used a trial and improvement approach and realised that the velocity must be $-3\mathbf{i} + 4\mathbf{j}$, which produced a speed of 5.

Q14.

This question was done well by many candidates. There were a few minor slips in part (a). Part (b) was a little more challenging. The most common errors were to omit one force, typically the weight leading to an answer of 160 N, or to make a sign error, for example $800g - T = 800 \times 0.2$ leading to a final answer of 7680 N.

Q15.

(a) was usually done correctly. Part (b) saw more errors from candidates, with some failing to subtract the 400 from the 2000 and using 2000 metres in their calculations. A few also

simply divided their distance by 18 rather than using a constant acceleration equation. In part (c), many candidates drew a graph with the correct shape, but many did not label the required speeds and times clearly on the axes. In many cases, a scale (for example 4, 8, 12, 16,...) had been put on the axis, rather than just indicating the initial and final speeds. Part (d) was generally done very well, but a few candidates did take the mean of 18 and 32 rather than using the total distance divided by the total time.

Q16.

This question was more demanding and candidates sometimes confused the horizontal and vertical aspects of the motion, for example by including the 250 horizontal component of velocity in an equation for the vertical motion.

However, part (a) was generally done well. In part (b), the confusion between the horizontal and vertical components of the motion was most apparent,

with expressions like $s = 250 \times 4 + \frac{1}{2} \times 9.8 \times 4^2$ being seen. Candidates who took this approach in part (b) then often made little progress in the later parts of the question. Similar problems were evident in part (c) when candidates tried to calculate the vertical component of the velocity. In part (d), some candidates used distances rather than components of the velocity to calculate the angle. In some cases there was a mixture of a velocity and a distance!

Q17.

In part (a), many candidates produced correct answers, but some stopped when they had obtained the velocity and did not go on to find the speed as requested. Part (b) was generally done well, but a few candidates did not give their final answer to the nearest degree. Part (c) was very much more demanding. A significant number of candidates mixed scalar and vector quantities in an attempt to answer the question. Statements like

$500 = \frac{1}{2} (0.5\mathbf{i} + 0.375\mathbf{j})^2$ were quite common and there were some similar ones that also included an initial velocity. The successful candidates used a variety of approaches, which can be seen in the mark scheme. The key difference between those who were successful and those who were not was the ability to connect the scalar distance of 500 to the position vector in a meaningful way.

Q18.

Most candidates showed that they knew the techniques involved in answering this question. Candidates generally answered part (a) correctly. Part (b)(i) was answered well, but in part (b)(ii) some candidates forgot to find the magnitude of $\mathbf{F}$, giving instead just the vector $\mathbf{F}$, $-40\mathbf{i} + 30\mathbf{j}$. Most candidates answered part (c) correctly. In part (d) the majority of candidates integrated to find the position vector, but a common error was to forget the $+\mathbf{c}$ term and thus to ignore the initial position vector. Some found the $+\mathbf{c}$ incorrectly, assuming that $\mathbf{c}$ was the initial position vector.

Q19.

There was a range of responses to this question, with candidates generally doing better on parts (a), (b) and (c) than on part (d).

In part (a) the most common error was to either include or omit a negative sign in the early

working, so that a negative time would have been produced if this had been followed through accurately.

In part (b) some candidates used an equation for the height rather than the velocity, but still equated it to zero. A few candidates ignored the initial height. Very few candidates realised that the ball would have the same speed when it returned to its initial position.

Q20.

Most candidates answered parts (a), (b) and (c) well, although some did get an answer of $-4.5j$ for part (b).

In part (c), the candidates who clearly stated and used a constant acceleration equation usually made a good start and completed the question, although there were a few arithmetic errors.

In part (e) many candidates did not attempt to find the magnitude of the force.

Part (f) was generally found more difficult with only a few good attempts. Several candidates did not attempt this part of the question.

Q21.

The candidates seemed to be helped by the printed answer in part (a), but some obtained

the printed result from the use of equations such as $5 = \frac{1}{2}(-9.8)t^2$. There were also some candidates who did not justify all of their working. There were a number of candidates who found it difficult to work with the two components of the velocity in parts (b), (c) and (d).

A reasonably common error was to use 5 instead of 15 in part (b) and then 15 instead of 5 in part (c). Some candidates also stopped after obtaining the vertical component in part (c) and did not go on to find the speed. Those who did find the components correctly were usually able to go on to find the angle correctly.

Q22.

Generally this question was answered very well. The most common difficulties were with

part (c). One error that appeared here was to use the equation $s = \frac{1}{2}(u + v)t$ with either u or v taken as zero. Often candidates who made errors in part (c) were able to gain the follow through marks in part (d).

Q23.

Most candidates made good attempts at this question. In part (b) (ii) there were some solutions that did not contain enough working to justify the award of marks, for example a

$= \frac{4}{0.5} = 8$ with no explanation of where the 4 came from. Some also simply found a way to get 8, for example from something like $9.8 - 2 \times 0.9 = 8$.

For part (b) (iii) there were many correct solutions, but some candidates took the acceleration as 9.8 instead of 8. The final part proved to be more difficult for the

candidates. While there were many candidates who produced good answers, there were also quite a lot who could not give a clear or relevant explanation.

Q24.

Parts (a) and (b) of this question were done very well and many candidates gained all four marks.

Part (c) proved to be more challenging. In part (c)(i), many candidates did not show how they started their working. An equation based on $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ was expected, but not always seen. Some candidates obtained the printed answer, but did not gain full marks as their starting point was not clear. Having both the acceleration and the initial velocity given in the question enabled many candidates to write down a correct expression for the velocity of the particle. Part (c)(iii) was found to be very challenging. If candidates were able to write down an expression for the speed of the particle they would often go on to find the times correctly. It was this vital first step that many candidates were unable to take. A few candidates made errors solving the quadratic equation once they had obtained it correctly.

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A reasonably common error was to use 5 instead of 15 in part (b) and then 15 instead of 5 in part (c). Some candidates also stopped after obtaining the vertical component in part (c) and did not go on to find the speed. Those who did find the components correctly were usually able to go on to find the angle correctly. Many candidates gained the two marks in part (e), sometimes without gaining any others in this question.

Q26.

The candidates generally did well with parts (a) and (b), which were fairly standard, but had more difficulties with the later parts. The main reasons that candidates lost marks in part (a) were because they made errors with the signs in the equations or because they did not form an equation for each particle. Those with sign errors in part (a) often did not obtain the correct tension in part (b).

In part (c), many errors were due to the candidates using the wrong values in the wrong places. For example, in part (c)(i), some used an acceleration of 9.8 instead of 1.96 while in part (c)(ii), some used an acceleration of 1.96 instead of 9.8. Similar confusion concerned the use of the distance of 4 metres in part (c)(i) instead of in part (c)(ii), while the time of 2 seconds was sometimes used in part (c)(ii) instead of in part (c)(i).

Q27.

The majority of the candidates were able to find the initial velocity as requested in part (a), but most found the rest of the question difficult. A reasonable number of the candidates were able to find the acceleration, some by calculating two velocities and using $\mathbf{v} = \mathbf{u} + \mathbf{a}t$, but others clearly could not find a strategy to find the acceleration. Part (c) was found to be more demanding. Some candidates did use the components correctly, but quite a

number of candidates did not recognise the approach that was required.

Q28.

Very many candidates were able to prove the result in part (a). Most considered the vertical component of the velocity and found the result easily. Some tried to find the time of flight first. In some cases these candidates halved their answer, but some did not do this or did not make it clear. A very small number of candidates used equations that they had learned for the time of flight or similar, but did not justify these results and so did not gain the marks. Candidates should not use learned formulae for the range etc in questions like this.

There were many different approaches to part (b)(i). Quite often candidates could produce a correct equation as a starting point, but failed to simplify it correctly. Often the candidates ended up with a mixture of $\sin \alpha$ and $\sin^2 \alpha$ terms. There were many minor arithmetic or algebraic errors. In part (b)(ii), many candidates used the incorrect time: their

approach was based on a time of $\frac{3 \sin \alpha}{2}$ instead of $3 \sin \alpha$.

Q29.

This question was done well by the vast majority of candidates, with many scoring full marks. Although not made often, the most common error was failing to multiply by 0.5, or divide by 2, when finding the distances in parts (b) and (c). Most candidates used a correct approach to find the average speed in part (d). Another error, seen in part (e), was to find 2400 but not subtract the answer to part (c).

Q30.

The candidates generally did well with parts (a) and (b), which were fairly standard, but had more difficulties with the later parts. The main reasons that candidates lost marks in part (a) were because they made errors with the signs in the equations or because they did not form an equation for each particle. Those with sign errors in part (a) often did not obtain the correct tension in part (b).

In part (c), many errors were due to the candidates using the wrong values in the wrong places.

For example, in part (c)(i), some used an acceleration of 9.8 instead of 1.96, while in parts (c)(ii) and (c)(iii), some used an acceleration of 1.96 instead of 9.8. Similar confusion concerned the use of the distance of 4 metres in part (c)(i) instead of the later parts, while the time of 2 seconds was sometimes used in the later parts instead of in part (c)(i).

In part (c)(iii), the most common approaches involved the use of constant acceleration equations. While some candidates used the correct numerical values in their equations, many made errors with the signs. Another common approach was to find two times and add them together. With this method also, sign errors were often made by the candidates.

Q31.

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approach was based on a time of $\frac{3 \sin \alpha}{2}$ instead of $3 \sin \alpha$.

Part (c) was often done well, but quite a few candidates incorrectly talked about the projectile having no mass.

Q32.

Virtually all candidates answered part (a) correctly. The common problem in part (b) was

in the integration of $400 \cos \frac{\pi}{2} t$. Apart from this, answers to part (b) were good, and most correctly found $t = 2$ in part (c). The majority of candidates also found correctly that the speed of the particle was 3, with only a few finishing with the velocity of the particle being -3 or $-3i$.

Q33.

There were many good solutions to part (a). Only a very small number of candidates ignored the instruction to form an equation of motion for each particle. A few candidates assumed that the heavier particle would accelerate upwards when released. Some of these candidates worked through to produce a negative acceleration and stated that magnitude would be the required value. However, some of the candidates who took this approach "lost" a negative sign in their working and ended up with a positive acceleration.

In part (b) there was one common error, which was seen on a fairly large number of scripts. This was to obtain the distance moved by each particle, but not to double this to give the distance between the particles. A very small number of candidates used an acceleration of 9.8 instead of 0.98.

Q34.

This question also provided more challenge for the candidates. In this question a number of candidates did assume that the acceleration due to gravity was positive.

In part (a), quite a number of candidates found the time of flight and then used half of this value to calculate the maximum height. There were however a number of cases where the candidates did not half their value for the time of flight and so produced solutions which did not relate to the maximum height.

Part (b) was much more demanding and relatively few correct solutions were seen. Those candidates who identified the required method were able to do well, but the majority made no attempt or produced working which could not lead to the required answer. Similarly part (c) was found to be quite difficult, but some candidates did use the answer from part (b) to put together correct solutions for part (c) when they had been unable to do part (b).

Q35.

Parts (a) and (b), about the assumptions, were generally answered well. The most common incorrect answer was to say that the peg was fixed in part (a).

There were many good solutions to part (c). Only a very small number of candidates ignored the instruction to form an equation of motion for each particle. A few candidates assumed that the heavier particle would accelerate upwards when released. Some of these candidates worked through to produce a negative acceleration and stated that magnitude would be the required value. However, some of the candidates who took this approach “lost” a negative sign in their working and ended up with a positive acceleration.

In part (d) there was one common error, which was seen on a fairly large number of scripts. This was to obtain the distance moved by each particle, but not to double this to give the distance between the particles. A very small number of candidates used an acceleration of 9.8 instead of 0.98.



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